

AWS HTTP Error Codes - Support Engineer Reference

Golden Rule: 4xx = Client / Config Problem | 5xx = Server / Infrastructure Problem (YOU OWN THESE)

No HTTP Code - Connection Timeout / Refused

Traffic is blocked at the network level before any HTTP response can be generated.

AWS Service	What to Look For
Security Groups	Inbound rules on the ALB SG - is port 80/443 open to 0.0.0.0/0?
Network ACLs (NACLs)	Stateless rules on the subnet - BOTH inbound AND outbound rules required
Route Tables	Is there a route to an Internet Gateway (IGW) for public subnets?
Internet Gateway	Is an IGW attached to the VPC?
NAT Gateway	For private subnets - is outbound traffic routed through a NAT GW?

VPC -> Security Groups -> Inbound Rules (look for port 80/443)

VPC -> Network ACLs -> check both inbound + outbound

VPC -> Route Tables -> confirm 0.0.0.0/0 -> igw-xxxxxxx

3xx - Redirects (Usually Not Errors)

Code	Meaning	AWS Service	What to Check
301/302	Redirect	ALB, CloudFront	ALB listener rules redirecting HTTP -> HTTPS; CloudFront default root object
304	Not Modified	CloudFront, S3	Cache headers (ETag, Cache-Control) - usually fine

4xx - Client / Configuration Errors

400 - Bad Request

Service	What to Check
API Gateway	Malformed JSON body, missing required headers, wrong Content-Type
ALB	Oversized headers exceeding ALB limits
WAF	Request blocked by AWS WAF rule - check WAF logs in CloudWatch

401 - Unauthorized

Service	What to Check
API Gateway	Cognito authorizer - token expired or missing Authorization header
API Gateway	Lambda authorizer returning Unauthorized policy
IAM	Missing execute-api:Invoke permission on the caller's IAM role

API Gateway -> Authorizers -> check authorizer type and token source

CloudWatch Logs -> API Gateway execution logs -> look for 'Unauthorized'

403 - Forbidden

Service	What to Check
S3	Bucket policy denying access; Block Public Access enabled; missing bucket ACL
CloudFront	WAF web ACL blocking request; geo-restriction rules; signed URL/cookie required
API Gateway	Resource policy denying source IP or VPC; IAM auth enabled but caller lacks permission
IAM	Caller's role/policy missing s3:GetObject, execute-api:Invoke, etc.
ALB	Listener rule configured to return a fixed 403 response

S3 -> Bucket -> Permissions -> Block Public Access + Bucket Policy

CloudFront -> Distribution -> Security -> WAF + Geo Restrictions

IAM -> Policy Simulator -> test the specific action being denied

Note: A 403 from S3 often means the bucket exists but access is denied - not that the file is missing (that's a 404).

404 - Not Found

Service	What to Check
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ALB	Listener rules don't match the request path - no rule returns a valid target group
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API Gateway	Wrong stage name in URL, resource path doesn't exist, method not defined
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S3 (static site)	Wrong object key, wrong bucket region endpoint, index document not configured
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CloudFront	Origin path misconfiguration; S3 key prefix mismatch
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Route 53	DNS pointing to wrong resource or stale record
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ALB -> Listeners -> View/Edit Rules -> check path-based routing

API Gateway -> Stages -> confirm stage name matches URL

S3 -> Objects -> confirm the exact key exists

429 - Too Many Requests

Service	What to Check
API Gateway	Usage plan throttle limits (requests/sec and burst); stage-level throttling
Lambda	Concurrency limit reached - check reserved vs. account-level concurrency
WAF	Rate-based rule triggered - check WAF sampled requests in console

API Gateway -> Usage Plans -> check throttle rate/burst

Lambda -> Configuration -> Concurrency -> check reserved concurrency

WAF -> Web ACLs -> Sampled Requests -> look for rate-based rule matches

5xx - Server / Infrastructure Errors (YOU OWN THESE)**500 - Internal Server Error**

Application crashed or threw an unhandled exception.

Service	What to Check
Lambda	Unhandled exception in function code - check CloudWatch Logs for ERROR
EC2	Application process crashed - check app logs via CloudWatch or SSH
ECS / Fargate	Container exiting with non-zero exit code - check ECS task stopped reason
API Gateway	Lambda integration returning malformed response (missing statusCode field)

CloudWatch -> Log Groups -> /aws/lambda/<function-name> -> filter for ERROR

ECS -> Clusters -> Tasks -> Stopped -> check 'Stopped Reason'

502 - Bad Gateway

The load balancer or proxy received an invalid response from the backend.

Service	What to Check
ALB	EC2 SG blocking inbound traffic from the ALB SG on the app port
ALB	Target group targets showing unhealthy; app not listening on correct port
API Gateway	Lambda returned a response missing required fields (statusCode, body)
CloudFront	Origin (ALB, S3, EC2) returned invalid response or is unreachable

ALB -> Target Groups -> Targets -> check health status + health check port

EC2 -> Security Groups -> confirm inbound rule allows ALB SG on app port (e.g. 8080)

CloudWatch -> ALB Access Logs -> look for backend response codes

Note: Most common root cause: EC2/ECS Security Group is missing an inbound rule that allows traffic FROM the ALB's security group.

503 - Service Unavailable

No healthy targets are available to handle the request.

Service	What to Check
ALB	All targets in the target group are failing health checks
Auto Scaling	Desired capacity is 0 or instances haven't launched/passed status checks yet
ECS	No running tasks in the service; task definition errors preventing launch
EKS	No ready pods available; node group scaled to 0

ALB -> Target Groups -> Targets -> all showing 'unhealthy'

EC2 -> Auto Scaling Groups -> check desired / min / max capacity

ECS -> Services -> check running task count vs. desired count

CloudWatch -> ALB Metrics -> HealthyHostCount = 0 is your signal

Note: Health check port blocked by a Security Group is the #1 cause of 503 on ALB.

504 - Gateway Timeout

The load balancer timed out waiting for the backend to respond.

Service	What to Check
ALB	ALB idle timeout (default 60s) - target took too long to respond
API Gateway	Hard 29-second maximum integration timeout - Lambda or backend too slow
Lambda	Function timeout setting too low; function hitting configured max duration
RDS / Aurora	Slow query causing the application to hang waiting on the database
ECS / EC2	Application under high CPU/memory load - check CloudWatch metrics

ALB -> Attributes -> Idle Timeout (increase if needed)

Lambda -> Configuration -> General -> Timeout setting

RDS -> Performance Insights -> identify slow queries

CloudWatch -> EC2/ECS metrics -> CPUUtilization, MemoryUtilization

Note: API Gateway's 29-second limit is hard and cannot be increased. If Lambda regularly hits it, consider an async pattern with SQS/SNS.

Layer -> Error Code Quick Reference

Layer	Typical Error	AWS Service
DNS	No response	Route 53
Network block	Connection timeout	VPC, Security Groups, NACLs
Load balancer -> EC2 blocked	502, 503	ALB, Security Groups
No healthy targets	503	ALB Target Groups, Auto Scaling
Backend slow	504	Lambda, RDS, EC2, ECS
App crash	500	Lambda (CloudWatch Logs), ECS
Auth / permissions	401, 403	IAM, Cognito, API Gateway
Rate limiting	429	API Gateway Usage Plans, WAF
Wrong path / resource	404	API Gateway, ALB Listener Rules, S3

First 5 Things to Check on Any 5xx

1. CloudWatch Logs

/aws/lambda/<name>, ECS task logs, or ALB access logs

2. ALB Target Group health

Are targets healthy? What port is the health check using?

3. Security Groups

Does the EC2/ECS SG allow inbound FROM the ALB's security group?

4. Auto Scaling / ECS desired count

Is anything actually running? Check desired vs. running count.

5. CloudWatch Metrics

CPUUtilization, HealthyHostCount, Lambda Errors/Duration